Nowcasting Recession Risk in the Euro Area: A Hierarchical Bayesian Approach

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Abstract

We propose a hierarchical Bayesian logit model for real-time nowcasting of recession risk in the Euro Area and its four largest member states. Building on Furno (2024), we combine the Economic Sentiment Indicator (ESI) and the Composite Indicator of Systemic Stress (CISS) within a multilevel structure that pools information across geographies. The hierarchical specification induces partial pooling, borrowing strength from the aggregate to stabilize country-level estimates—particularly for economies with limited recession history under the single currency. Using quarterly data from Eurostat and the ECB, we estimate the model with a five-year out-of-sample window. Our results show that the model accurately identifies all major EA recessions in sample and captures the divergence in recession risk across member states in recent years, with Germany exhibiting persistently elevated probabilities relative to Spain. A pseudo-real-time evaluation on historical GDP vintages shows that real-time nowcasts remain informative about the contemporaneous recession state, and a head-to-head comparison against a PMI-based specification assesses which sentiment proxy is better suited for monitoring going forward.

1 Introduction

Official business cycle dating in the Euro Area, as carried out by the Centre for Economic Policy Research (CEPR), is inherently retrospective: recessions are announced only months after their onset. For policymakers and financial market participants who require timely assessments of macroeconomic conditions, such delays are costly. This has motivated a substantial body of research on real-time monitoring of economic activity, which broadly falls into two strands. The first, pioneered by Giannone et al. (2008) and extended by Bańbura et al. (2013), uses high-frequency indicators to track current-quarter GDP growth. The second, rooted in the work of Estrella & Mishkin (1998) and Chauvet & Piger (2008), models the binary recession state directly via probit or logit specifications.

Furno & Giannone (2024) propose a logit-based framework for nowcasting recession risk that combines sentiment and financial-stress indicators. A natural limitation of estimating such models on a country-by-country basis is that individual Euro Area member states—particularly those that adopted the euro relatively recently—may have experienced only a small number of recession episodes, leading to imprecise and unstable coefficient estimates. We address this issue by embedding the recession logit within a hierarchical Bayesian structure that allows country-specific parameters to deviate from a common distribution, thereby inducing partial pooling across geographies.

Our approach also draws on the "Growth-at-Risk" framework of Adrian et al. (2019), who emphasize the role of financial conditions in shaping the distribution of future output growth. We incorporate the Composite Indicator of Systemic Stress (CISS) of Holló et al. (2012) as a measure of financial fragility and the Economic Sentiment Indicator (ESI) as a gauge of macroeconomic expectations. The resulting model produces real-time recession probability estimates for the Euro Area aggregate and four major member states (Germany, France, Italy, and Spain), along with Bayesian credible intervals that quantify estimation uncertainty.

2 Data

We construct a quarterly panel covering the Euro Area aggregate and its four largest economies: Germany (DE), France (FR), Italy (IT), and Spain (ES). All series are obtained from Eurostat and the ECB Statistical Data Warehouse and are aggregated from monthly to quarterly frequency by simple averaging.

The dependent variable, $y_{i,t}$, is a binary technical recession indicator equal to one when quarter-on-quarter real GDP growth is negative in two consecutive quarters ($\Delta\text{GDP}_{i,t} < 0$ and $\Delta\text{GDP}_{i,t-1} < 0$). Real GDP data are drawn from Eurostat table `namq_10_gdp` (chain-linked volumes, percentage change on the previous quarter). The two predictors are the Economic Sentiment Indicator (Eurostat table `ei_bssi_m_r2`), which summarizes sectoral survey-based expectations, and the CISS (ECB key `CISS.D..Z0Z.4F.EC.SS_CIN.IDX`), which captures systemic stress across financial markets. Both predictors are standardized to zero mean and unit variance prior to estimation.

3 Methodology

We specify a hierarchical Bayesian logit model in which country-specific coefficients are drawn from a common prior distribution. Let $y_{i,t} \in \{0, 1\}$ denote the recession indicator for geography i in quarter t. The probability of recession is modeled as:

$$\text{logit}\big(P(y_{i,t} = 1)\big) = \alpha_i + \beta_{1,i}\,\text{CISS}_{i,t} + \beta_{2,i}\,\text{ESI}_{i,t},$$

where the country-specific parameters are decomposed into population-level means and group-level deviations:

$$\alpha_i = \alpha_0 + u_{0,i}, \qquad \beta_{1,i} = \beta_1 + u_{1,i}, \qquad \beta_{2,i} = \beta_2 + u_{2,i}.$$

The vector of random deviations $\mathbf{u}_i = (u_{0,i},\, u_{1,i},\, u_{2,i})'$ is assumed to follow a multivariate normal distribution with mean zero and an unstructured covariance matrix Σ. This formulation nests two polar cases: when $\Sigma \to \mathbf{0}$, the model collapses to a pooled logit with common coefficients; when $\Sigma \to \infty$, it reduces to separate country-level estimates with no pooling. The intermediate case induces shrinkage, pulling country estimates toward the Euro Area mean while preserving cross-country heterogeneity.

The model is estimated via Hamiltonian Monte Carlo using the `brms` package (Bürkner, 2017) in R, with two chains of 2,000 iterations (1,000 warmup each). We specify weakly informative priors: $\alpha_0 \sim \mathcal{N}(0, 5)$ and $\beta_1, \beta_2 \sim \mathcal{N}(0, 2)$ on the logit scale; the group-level standard deviations and correlations keep the `brms` defaults (half-Student-$t(3,\, 0,\, 2.5)$ and LKJ(1), respectively). The training sample excludes the most recent five years of data, which are reserved for out-of-sample inference. Predictors are standardized using the mean and standard deviation of the training window only, so that no post-cutoff information enters the out-of-sample estimates.

Model artifacts not found at dashboard/data/probabilities_esi.csv. Run new_code/update_dashboard_data.R on a data host, commit dashboard/data/, and re-render. The results tables and figures below are omitted.

4 Results

The posterior estimates of the population-level effects indicate that both predictors carry the expected signs and are statistically credible. The CISS coefficient is positive, implying that elevated systemic financial stress increases the probability of recession, while the ESI coefficient is negative, indicating that improving sentiment reduces recession risk. Table **??** reports the posterior means and 95% credible intervals for the fixed effects.

?@fig-recession-probabilities displays the estimated recession probabilities for the Euro Area aggregate and the four member states, together with 95% credible intervals. Grey shaded bars mark historical technical

recessions, and the vertical dashed line separates the in-sample estimation period from the out-of-sample inference window (the most recent five years).

In sample, the model successfully identifies all major Euro Area recessions—the Global Financial Crisis of 2008–2009, the Sovereign Debt Crisis of 2011–2013, and the COVID-19 contraction of 2020—with sharp probability spikes exceeding 80%. The out-of-sample period, during which no parameter updating occurs, reveals a pronounced divergence across member states since 2022. Germany displays a sustained increase in recession probability, consistent with the structural challenges facing its manufacturing sector, whereas Spain maintains comparatively lower risk estimates throughout the same period.

A notable feature of the results is the tightness of the credible intervals at the country level, despite the relatively small number of recession episodes available for individual member states. This is a direct consequence of the hierarchical structure: by pooling information across geographies, the model regularizes country-specific estimates and avoids the volatile, extreme probabilities that would arise from fitting each country independently.

5 Real-time evaluation

The estimates above use the currently published data history, which is itself the product of revisions. To assess how the model performs on the information actually available at each point in time, we run a **pseudo-real-time evaluation**. The exercise reconstructs, for each month since 2015, the GDP panel as it was known on that date: Euro Area aggregates come from the Eurostat quarterly GDP vintage archive (`ei_na_q_vtg`), and country-level vintages from the OECD "Short-term economic statistics revisions" archive, which preserves monthly snapshots of the MEI database since 1999. Predictors enter only once published—the ESI is assumed available on day 29 of its reference month, the composite PMI on day 26—and a quarter's predictor value is formed only when all three months are published. At each event date on which at least one area's latest available GDP quarter changes, the hierarchical model is re-estimated on the pooled panel with the same five-year training gap, and nowcasts the recession probability of each area's latest GDP quarter. Each (area, quarter) is scored once, at its first nowcast opportunity, using the Brier score, the log score, and AUC; paired comparisons use Diebold–Mariano tests with a Newey–West HAC variance and the Harvey et al. (1997) small-sample correction.

As a first validation, Table 1 compares, for the Euro Area aggregate, the real-time nowcasts against the same model evaluated on the latest (fully revised) vintage. Real-time probabilities are naturally less accurate than the ex-post-vintage benchmark—GDP data are revised substantially around turning points—but the degradation is moderate, and the real-time series remains informative about the recession state.

Table 1: Euro Area aggregate: pseudo-real-time nowcasts versus latest-vintage evaluation (CISS + ESI specification; 47 evaluation events, 2014-Q3 to 2026-Q1).

Comparison	N	Brier	Log score	AUC
Real-time probability (real-time vintage)	47	0.024	0.122	0.767
Real-time probability (latest vintage)	47	0.045	0.204	0.689
Latest-vintage in-sample probability (latest vintage)	113	0.059	0.202	0.906
Latest-vintage probability on the common real-time sample	47	0.043	0.177	0.742

Because the sentiment-based predictor is not obviously optimal—the composite PMI is available earlier in the month and is never revised, while the ESI is broader but published only at month-end—we also run a **head-to-head comparison** of the two specifications (CISS + ESI versus CISS + PMI) in the same pseudo-real-time design, across the Euro Area and the four member states, with Diebold–Mariano tests of equal predictive ability on paired loss differentials.

The ESI-vs-PMI head-to-head run is pending: it requires the licensed S&P Global PMI workbook and is executed in the production environment (see new_code/PROD_RUN_ESI_VS_PMI.md). Results will be inserted here once the run completes.

Three limitations of the real-time exercise deserve note. First, the ESI and CISS are available as full currently-published histories rather than true vintages, so they are real-time only with respect to publication timing; the PMI, which S&P Global does not revise, is treated exactly. Second, the OECD editions are monthly snapshots of the MEI database rather than national first releases, though they are the best publicly available archive of country GDP vintages. Third, the technical-recession label is itself vintage-dependent: scoring against the contemporaneous vintage, as done here, is the correct real-time criterion but differs from scoring against revised data.

6 Conclusion

This paper introduces a hierarchical Bayesian framework for nowcasting recession risk in the Euro Area that addresses the fundamental trade-off between exploiting cross-country commonality and preserving country-specific structure. The partial-pooling specification produces stable, well-calibrated probability estimates even for economies with limited recession histories. Our results demonstrate that combining a financial-stress indicator (CISS) with a sentiment measure (ESI) in such a framework delivers accurate in-sample recession detection and informative out-of-sample signals of divergent macroeconomic conditions across member states. A pseudo-real-time evaluation on historical GDP vintages confirms that these properties survive the data-revision process, and a head-to-head comparison against a PMI-based specification identifies which sentiment proxy is better suited for monitoring going forward.

The framework can be extended in several directions. Incorporating additional predictors—such as term-spread variables, loan growth, or survey-based inflation expectations—could further sharpen the nowcasts. The hierarchical structure could be enriched by allowing partial pooling at a sectoral level, thereby capturing within-country heterogeneity in recession dynamics. Finally, replacing the technical-recession definition with committee-based dating, or assessing calibration formally, would provide complementary robustness checks.

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